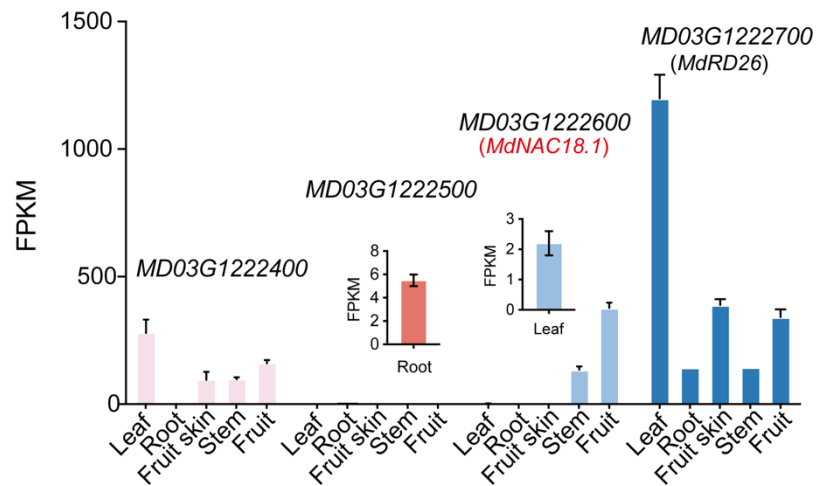
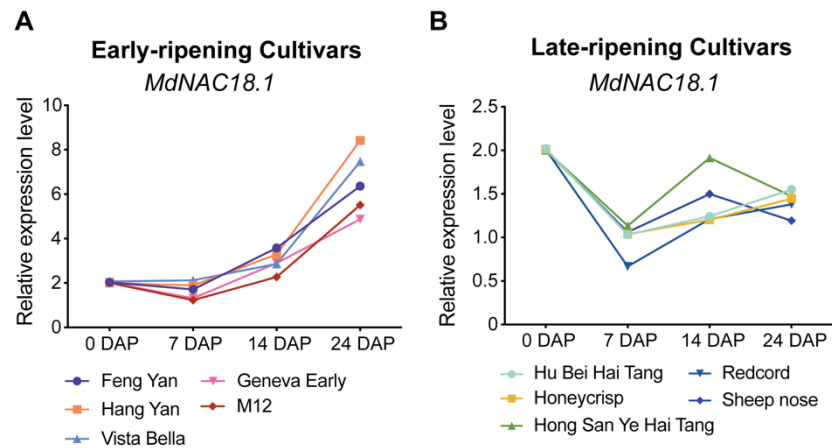




Supplementary Figure S1. Tissue-specific expression of the four candidate genes in *Malus fusca*. Error bars indicate SD ($n = 3$). Fragments Per Kilobase Million (FPKM) of the four candidate genes in different tissues were retrieved from the apple eFP browser (http://bar.utoronto.ca/~asher/efp_apple/cgi-bin/efpWeb.cgi).

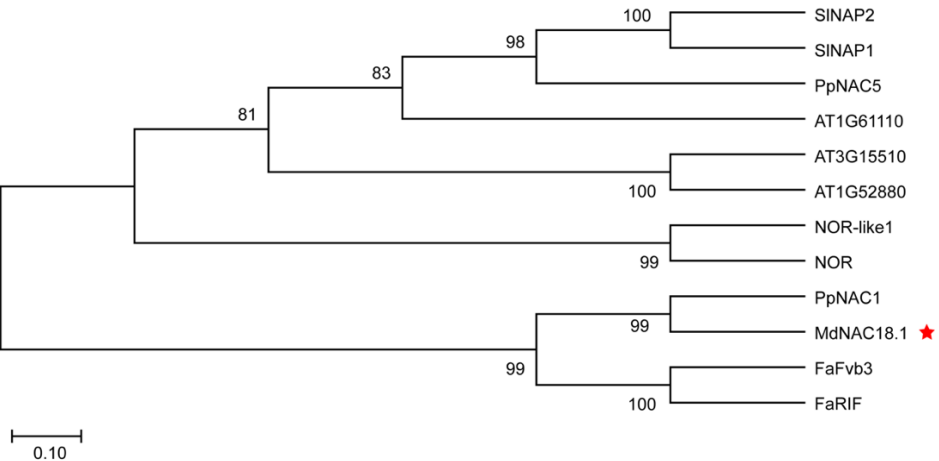


Supplementary Figure S2. Expression of *MdNAC18.1* in early- and late-ripening cultivars in the early stages of fruit development. Each sample has three biological replicates. DAP, days after pollination.

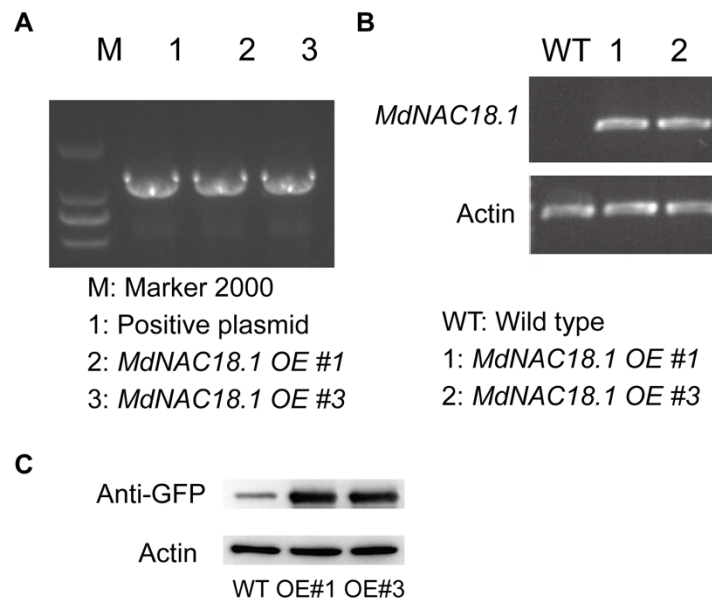
A

NAM domain

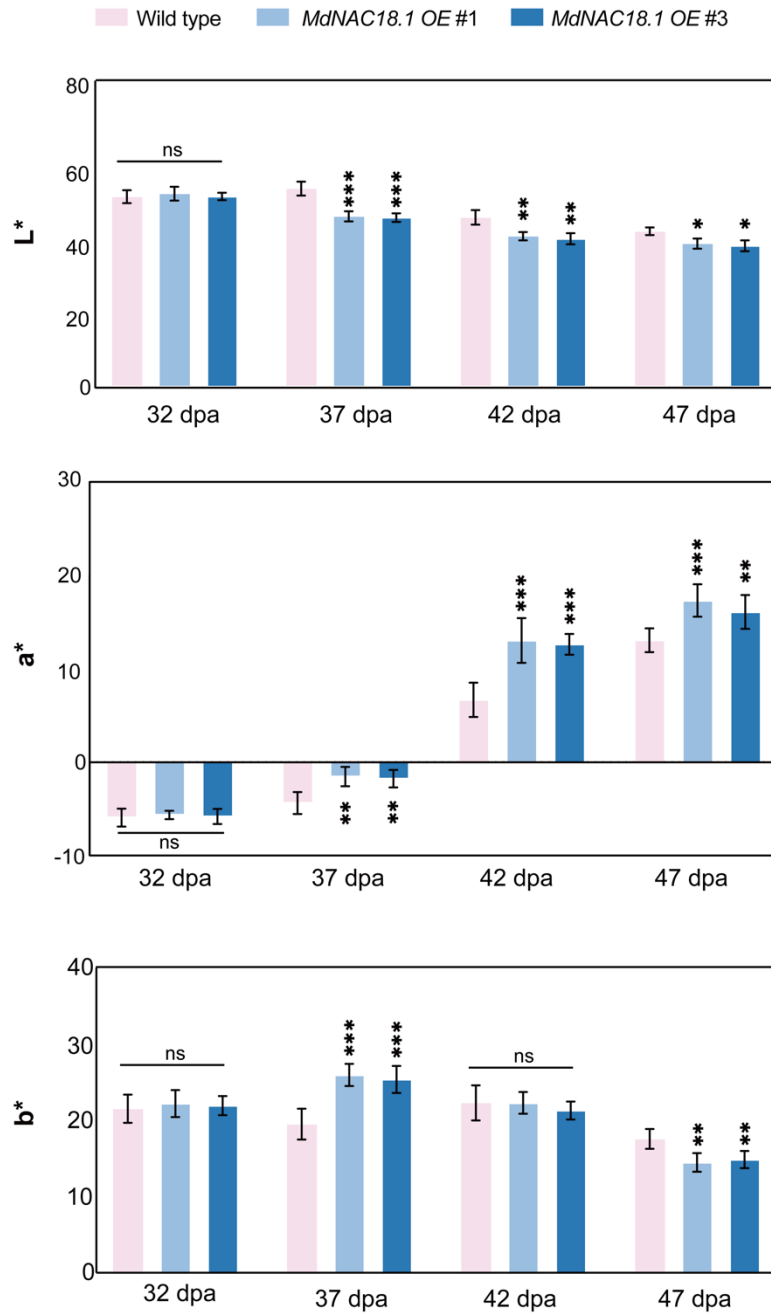
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AT1G61110	MENMGSSIGP-----GHPHLPFGGRFHPTDEELVVHYLKKKADSVLPVSIIEADLYKFDPWELPAKASFGEHEWYFFSPDRKYPNGVRPN 89
AT3G15510	MEST-DSSGGP-----PPQPPLPPGGRFHPTDEELVVHYLKKKAASAPLVAIIAEVDLYKFDPWELPAKASFGEQEWYFFSPDRKYPNGARPN 90
FaFvb3	MEST-DSSSSGQ-----PPQPPLPPGGRFHPTDEELVVHYLKKKASAPLPIAIIAEVDLYKFDPWELPAKATFGEQEWYFFSPDRKYPNGARPN 91
FaRIF	MEST-DSSSSGQ-----PPQPPLPPGGRFHPTDEELVVHYLKKKASAPLPIAIIAEVDLYKFDPWELPAKATFGEQEWYFFSPDRKYPNGARPN 91
MdNAC18.1	MECT-DSSAGSQQQHQQP-PPQPPLPPGGRFHPTDEELVVHYLKKKASAPLVAIIAEVDLYKFDPWELPAKATFGEQEWYFFSPDRKYPNGARPN 98
NOR-like1	MEST-DSSSTGSH-----HQPOLPPGGRFHPTDEELVVHYLKKKVASVLPVSIIEADLYKFDPWELPAKATFGEQEWYFFSPDRKYPNGARPN 89
NOR	MEST-DSSSTGTR-----HQPOLPPGGRFHPTDEELVVHYLKKRVAGAPIPVDIIGIDLYKFDPWELPAKATFGEQEWYFFSPDRKYPNGARPN 89
PpNAC1	MEST-DSSSTASQQQQQQPQP-PPQPPLPPGGRFHPTDEELVVHYLKKKVTSAPLVAIIAEIELYKFDPWELPAKATFGEQEWYFFSPDRKYPNGARPN 99
PpNAC5	-----MGVPETD-----PLSQLPPGGRFYPTDEELLVQLCRKAVAGYQFSLQIIAEIDLYKFDPWVLPKSAIFGEKEWYFFSPDRKYPNGSRPN 87
SINAP1	-----MVGK-----ISSDLPFGGRFHPTDEELIMYLYRQATSRPCPVSIIEIDVYKFDPWVLPKAEFGDNWYFFTPDRKYPNGVRPN 82
SINAP2	-----MVGKN-----NSNHLPPGGRFHPTDEELIMYLRNQATSKPCSSIIEVDVYKFDPWELPEKTEFGEKEWYFFTPDRKYPNGVRPN 83
Consensus	RAATSGYWKATGTDKPV-----GT-KVGVKKALVFY-GKPPKG-KTNWIMHEYRL-DNK--T-NK-PPGC-----KKNSLRDDDWVLCRIYKKN
AT1G52880	RAATSGYWKATGTDKPVIST---GGGSKKVGKVKALVFYSGKPPKGVKSDWIMHEYRLTDNK--PTHI---C-D--FGNKKNSLRDDDWVLCRIYKKN 178
AT1G61110	RAATSGYWKATGTDKPIFTC-----NSHKVGKVKALVFYGGKPPKGIKTDWIMHEYRLTDGNLSTAAK-PPDL-----TTRKKNSLRDDDWVLCRIYKKN 178
AT3G15510	RAATSGYWKATGTDKPVLAS-----DGNQKVGKVKALVFYSGKPPKGVKSDWIMHEYRLIENK--PNNR-PPGC-D--FGNKKNSLRDDDWVLCRIYKKN 179
FaFvb3	RAATSGYWKATGTDKPVLSSTDEGGGGRKVGKVKALVFYRGKPPKGIKTNWIMHEYRLADNN--TSNKPPPGCHD--LGSKKNSLRDDDWVLCRIYKKN 187
FaRIF	RAATSGYWKATGTDKPVLSSTDEGGGGRKVGKVKALVFYRGKPPKGIKTNWIMHEYRLADNN--TSNKPPPGCHD--LGSKKNSLRDDDWVLCRIYKKN 187
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NOR-like1	RAATSGYWKATGTDKPVLTSS---GDTQKVGKVKALVFYGGKPPKGIKTNWIMHEYRLADNK--TNNK-PPGC-D--LANKK-SLRDDDWVLCRIYKKN 177
NOR	RAATSGYWKATGTDKPVFTS-----GGTQKVGKVKALVFYGGKPPKGIKTNWIMHEYRVVENK--TNNK-PLGC-DNIVANKKGSRLDDDWVLCRIYKKN 180
PpNAC1	RAATSGYWKATGTDKPVLTSS---GGTQKVGKVKALVFYGGKPPKGIKTNWIMHEYRLADSK--TSNK-PPGC-D--LGNKKNSLRDDDWVLCRIYKKN 188
PpNAC5	RVAGSGYWKATGTDKIITTE-----GRKVGKVKALVFYVGKAPKGIKTNWIMHEYRLIEPS-----RKNSSKLDEWVLCRIYKKS 163
SINAP1	RAAVSGYWKATGTDKAIYSA-----NKYVGKVKALVFYGGKPPKGIKTDWIMHEYRLDSK--SQ-----TSKQSGSMRLDDDWVLCRIYKKN 162
SINAP2	RAAVSGYWKATGTDKGIYS-----GTKYVGKVKALVFYGGKPPKGIKTDWIMHEYRLSESR--TQPTRPNGS-----MRLLDDWVLCRIYKKN 163
Consensus	N--R-----D-----D-----P-S-----LP---S-S-----N-----D
AT1G52880	NSTASRHHHHLHHIHLNDNHHRHDMID--DDRFRHV-----PGLH-----FPA-----IFSDNNDPTA----IYD 234
AT1G61110	SSQRP-----THERVLLR-----EDLMGDM-----LSKSSANSSSTVLDNNDNNNNNEEHFFD 228
AT3G15510	NASR-----HVDNDKDHDMI--DYIFRKI-----PPSLMAAAGTGLQHQQHHNVSRSMNFFPGKFSGGGYS-IFSDGNGTS-----IYD 250
FaFvb3	NTHR-----PMD-QEE-SM-----DDMMGSS-----FPLSKLH-----R-----LPPKSTTTSTYG-QFM--DNYHN-----FYD 236
FaRIF	NTHR-----PMD-QEE-SM-----DDMMGSS-----FPLSKLH-----R-----LPPKSTTTSTYG-QFM--DNYHN-----FYD 236
MdNAC18.1	NTHR-----PMDHERED-SM-----EDIMGPL-----MPSISHVGHQNMK-----LHFPKSNSSYGPPFM--ENDQI-----FFD 246
NOR-like1	NTQR-----PIDHERDDLNI--DDMMGSSSIHSPICPNSMSPNIFGQPK-----IPQLKS-SNFGTTLTHDQNGN-----LYE 244
NOR	NTQR-----SIDDLHD-----MLGSI--PQVVPNST-----LQGIKP-SNYGTILL--ENESN-----MYD 226
PpNAC1	NSHR-----PMDLERED-SM-----EDMMGPL-----MPSISHVGHQNMN-----LHLPKSNTNYGPPFI--ENDQI-----IFD 247
PpNAC5	SSSAAQKPMTTSSVSKHSNGSSSSCSQLDDVLEWL-----PDIDDRCTF-----LPRINSKLT--LQQQQEDSKL--GFQ 231
SINAP1	NLGR-----TIEMMKVE-----EELEAQ--NVSTTNEIEVVGQPQTMK-----LPRICSLS--HLLELD-----YFG 217
SINAP2	NLER-----AIEMMKVE-----EDTQEPQIMSVTNPPIHEVVASNGQQLTK--LPRTCSLS--HLLEMD-----YFG 220
Consensus	-G-----KR-----YWN-----R-----S-
AT1G52880	GGGGGGYGGGSYMNHCFASS-----KQEQLFPVPMMTSLNQDSGI-----GSSSSPSKRFN-----GGGV-----GDC 294
AT1G61110	-GMVVS-----SDKRSLCGGYRMGEASGSS-----FGSFLSKRFH-----HTGD-----LNNNDNYNVSF 279
AT3G15510	GGGMINNIGTDSVDHNDNADVVGLNHASSSGPMMAANKRLT-PVPYWPVADEEQDA-----SPSKRFHGV-----GGGG-----GDC 322
FaFvb3	-GMVSSEGINTSASFPLNSGMASSSL-----PLKRG-LPNLYWNHDQDEAG-----LSRRLHLDNSE-STGN-----GSI 299
FaRIF	-GMVSSEGINTSASFPLNSGMASSSL-----PLKRG-LPNLYWNHDQDEAG-----LSRRLHLDNSE-STGN-----GSI 299
MdNAC18.1	-GMSSTDGSAS-----THQL-----QLKRIPVPSWYNNQEDDQTA--ADPSSSKRIQLHLQD-NGTNSNGAGTNGGSTNNNSSSI 319
NOR-like1	-G-----GSQY-----SSKRLP-ANLYWNDQGGASN-----DNSQSTKRFLENME-DGLN--MNA-----RADEQNGSI 300
NOR	-GIMNNTNDIIN-----NNRSIPQI-----SSKRTMHGGLYWNDEATTTTTIDRNHSPNTKRFLENNEEDGLN--MNNISRTNHEQSSI 308
PpNAC1	-GIMSTDGSASLS-----NGTSQL-----PLKRSIVPSLYWNDQEDDQTA--GASSSKRVVQLHLQD-SGTN-----NSVAANNNTSI 318
PpNAC5	MGSNFDWASLAGLVNVPCLCPNNQP-----QQSQGMNVNYSNNDMYVPSIPPLCHVESPPERLAKTVDEEVQSGF-----RTQRVDNSGFFQNS 317
SINAP1	-----SIPQLLSNDLLYDDGQYTMNN-----VNNTSNVDQVSSQQQNTNNIT-----SNN 262
SINAP2	-----SISQLFDNNSYNTISQNNTL-----
Consensus	---LS--PQ-----LG-----D-----FR---Y-QLPGMNWYS-----
AT1G52880	STMAATPL-----MQNQGGIY-----QLPGLNWYS-----320
AT1G61110	VSMLEIPQSSGFHA-----NGVMDTTSSLADHGVLR-QAF-QLPNMNWHS-----323
AT3G15510	SNMSSSMMEETPPL-----MQQGGVL--GDG-LFRITTSY-QLPGLNWYS-----364
FaFvb3	ATLLSQLPTPPPLH-----QQPMLGSLGGDA-LFRSTQY-QLPGMNWYS-----343
FaRIF	ATLLSQLPTPPPLH-----QQPMLGSLGGDA-LFRSTQY-QLPGMNWYS-----343
MdNAC18.1	ANLLSQLPT-PPLH-----QHAVLGSL--GDG-IFR-TPY-QLPGMNWYSESNLG 364
NOR-like1	VSLLS-----QQVVLGSL--SEG-VFR-TPY--SGMNWYS-----329
NOR	ANFLSQPQN-PSITQQQQQEEVLGSL--NDGVVFR-QPYNQYTPGMNWS-----355
PpNAC1	ANLLSQLPT-PPLH-----QHAMLGSL--GDG-LFR-TPY-QLPGMNWYSESNLG 363
PpNAC5	NVMTQNFNCPTDPYG-----YSNRLGRSGSLGFGGAEK-----349
SINAP1	CNIFNYQQP-----LFVNPTE-QSQ-----282
SINAP2	MTNVNGYVMP-----HQAMEKFLGGEVSQISMNPSY-QFQ-----275



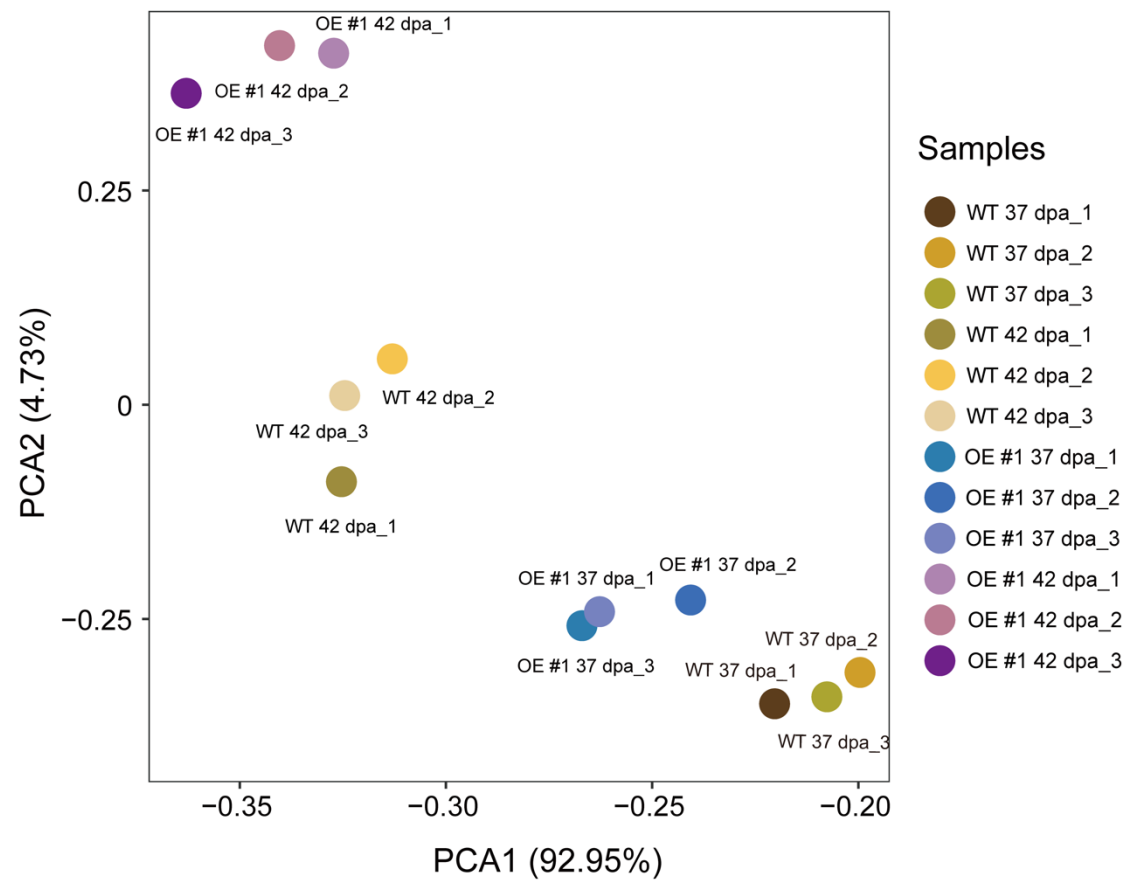
Supplementary Figure S3. Sequence alignment and phylogenetic analysis of MdAC18.1. **A)** Sequence alignment of NAC proteins. Conserved domains were indicated by black squares. The yellow box indicates the name of the protein sequence. Pink highlighting indicates the same amino acids in the protein sequence alignment. **B)** Phylogenetic analysis of MdNAC18.1 proteins and their homologs. Evolutionary distances were estimated with a neighbor-joining algorithm. The numbers above the branches indicate the percentage from 1,000 bootstrap. The scale bar indicates the average number of amino acid substitutions per site. Accession numbers are as follows: *Solanum lycopersicum* (SINAP2, Solyc04g005610; SINAP1, Solyc05g007770; NOR-like1, Solyc07g063420; NOR, Solyc10g006880), *Prunus persica* (PpNAC1, Prupe.4G187100; PpNAC5, Prupe.4G186800), *Fragaria ananassa* (FaRIF, FvesH4_3g20700; FaFvb3, FvH4_3g28570), *Malus domestica* (MdNAC18.1, Md03G1222600), and *Arabidopsis thaliana* (AT1G61110; AT3G15510; AT1G52880).



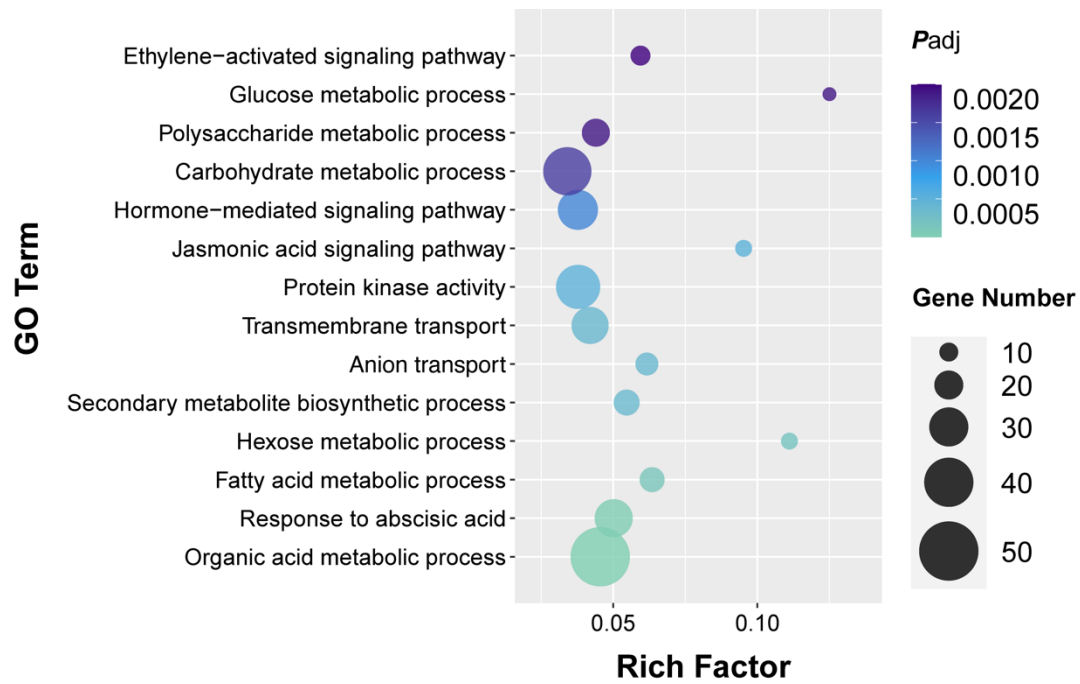
Supplementary Figure S4. Identification of the transgenic tomatoes overexpressing *MdNAC18.1* (OE). **A)** Identification of the transgenic tomatoes via PCR. Plasmid DNA was used as a positive control. **B)** Identification of the transgenic tomatoes via RT-PCR. **C)** Western blotting of *MdNAC18.1* protein in the transgenic tomatoes. The ripening fruits were used to extract DNA and total protein in A, B, and C, respectively. OE, overexpression.



Supplementary Figure S5. Color development in wild type and *MdNAC18.1* OE tomato fruits. Pericarp color (L^* , a^* , b^*) was measured with a handheld colorimeter using the CIE $L^*a^*b^*$ color system. L^* indicates lightness, a^* indicates red to green, and b^* indicates yellow to blue. The time of tomato fruits development is plotted on the X-axis, and the pericarp color (L^* , a^* , b^*) is plotted on the Y-axis. Error bars indicate SD ($n = 10$). Asterisks indicate the significant differences between the *MdNAC18.1* OE fruits and the wild type in each group (Student's two-tailed t test, ** $P < 0.01$, *** $P < 0.001$, ns, not significant). OE, overexpression. dpa, days post-anthesis.

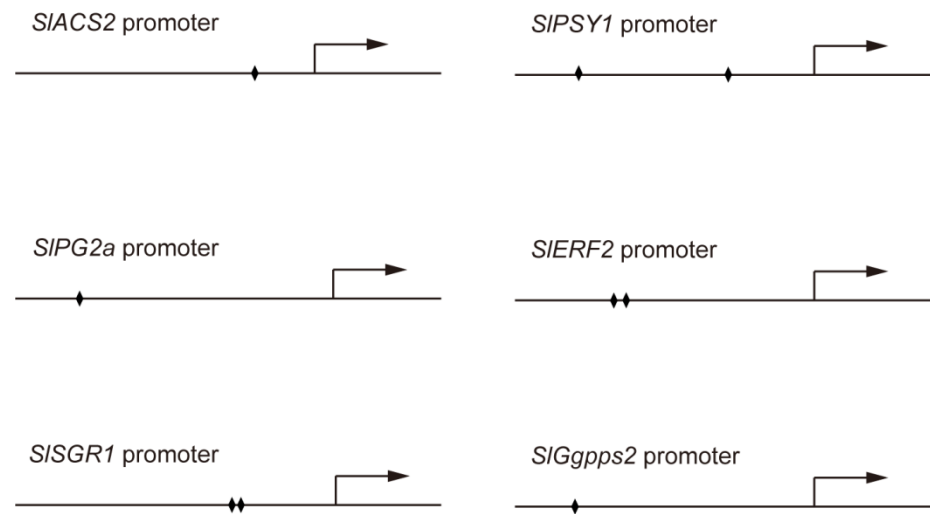


Supplementary Figure S6. Principal component analysis (PCA) of transcriptome data. Each sample has three biological replicates. WT, wild type; OE, overexpression. dpa, days post-anthesis.

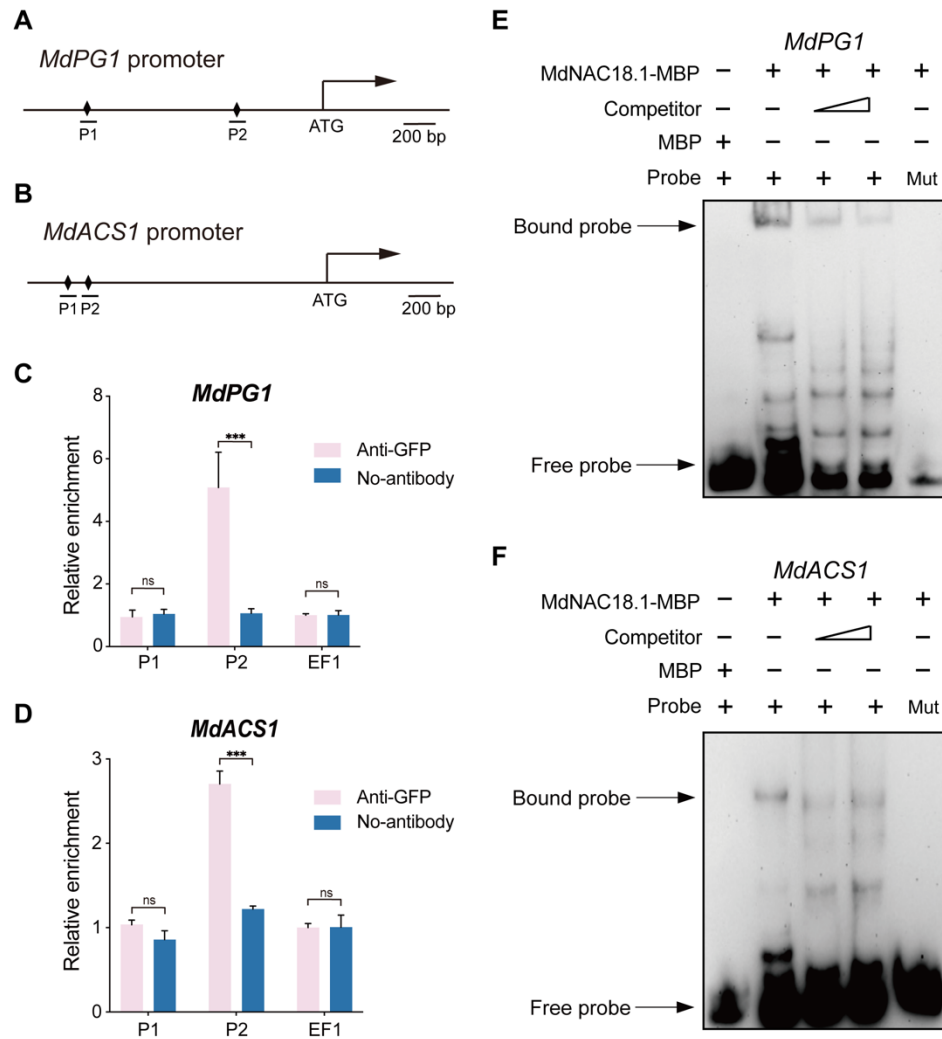


Supplementary Figure S7. GO enrichment analysis of DEGs between wild type and *MdNAC18.1* *OE* fruits. Results of classification within the molecular function category with *p* values from the statistical over representation test are shown.

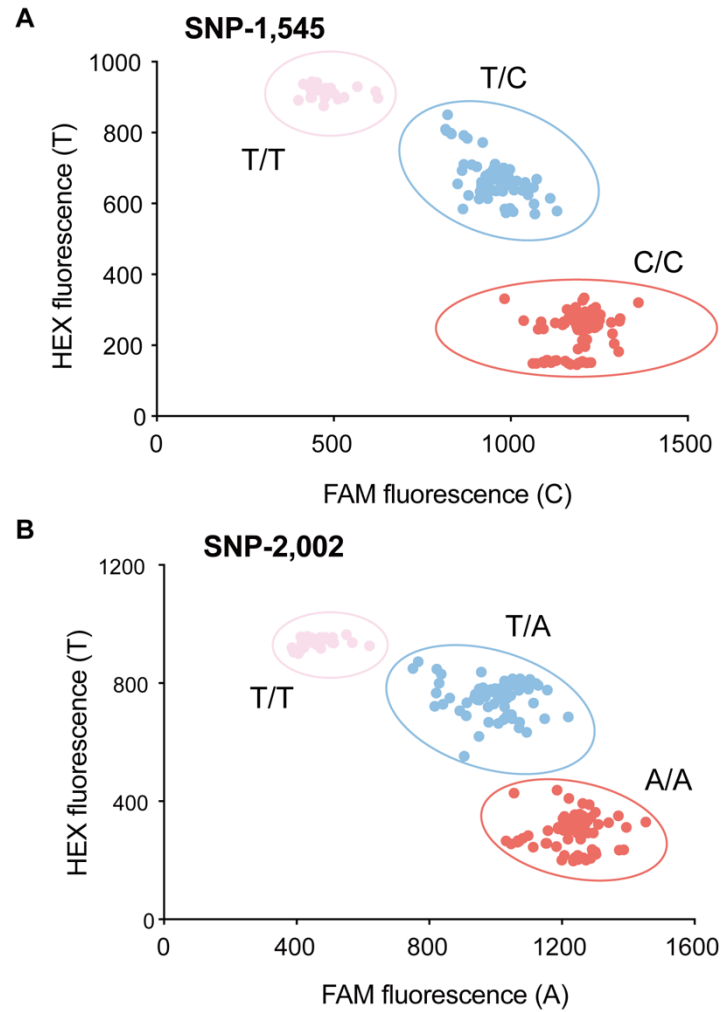
♦ NACRS motif — 200 bp



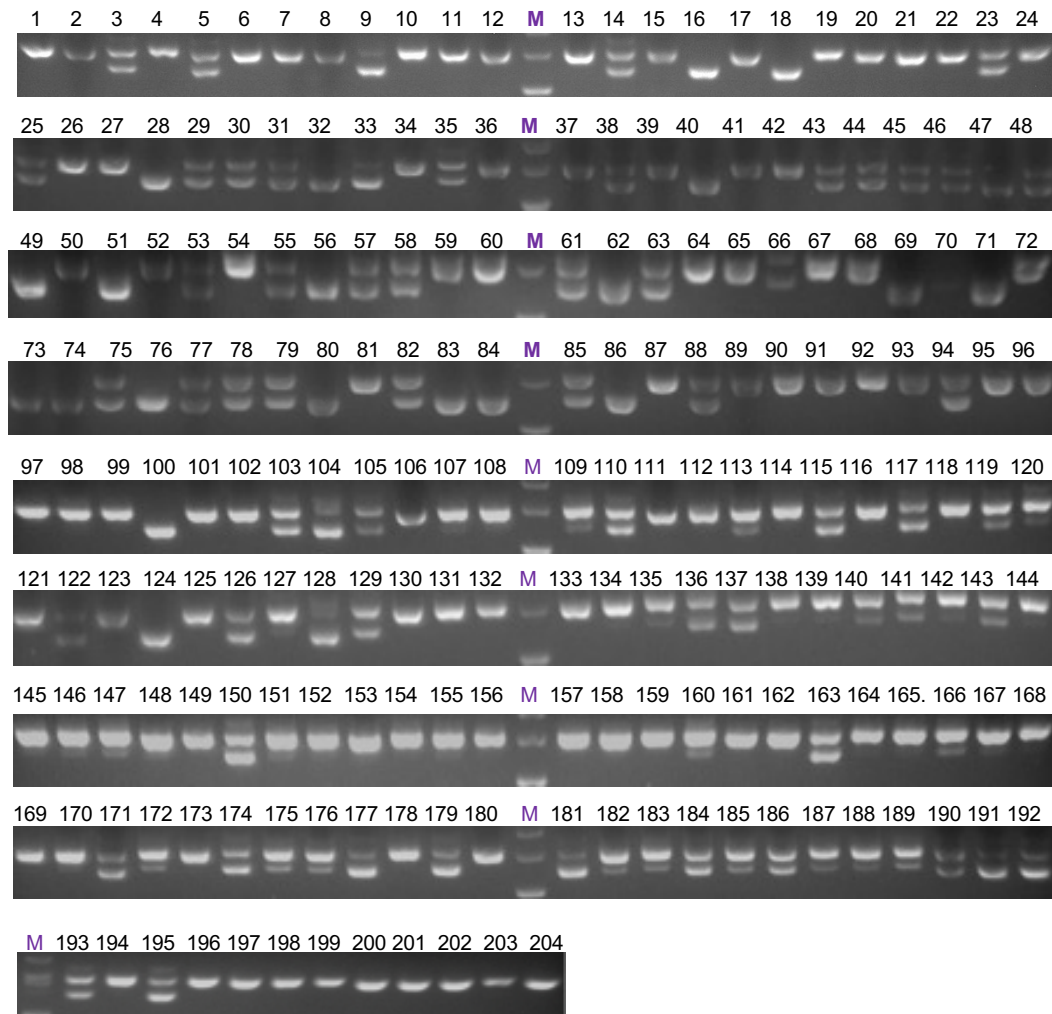
Supplementary Figure S8. Analysis of the promoters of *SIACS2*, *SIPSY1*, *SIPG2a*, *SIERF2*, *SISGR1*, and *SIGpps2*. The black arrow represents the transcription start.



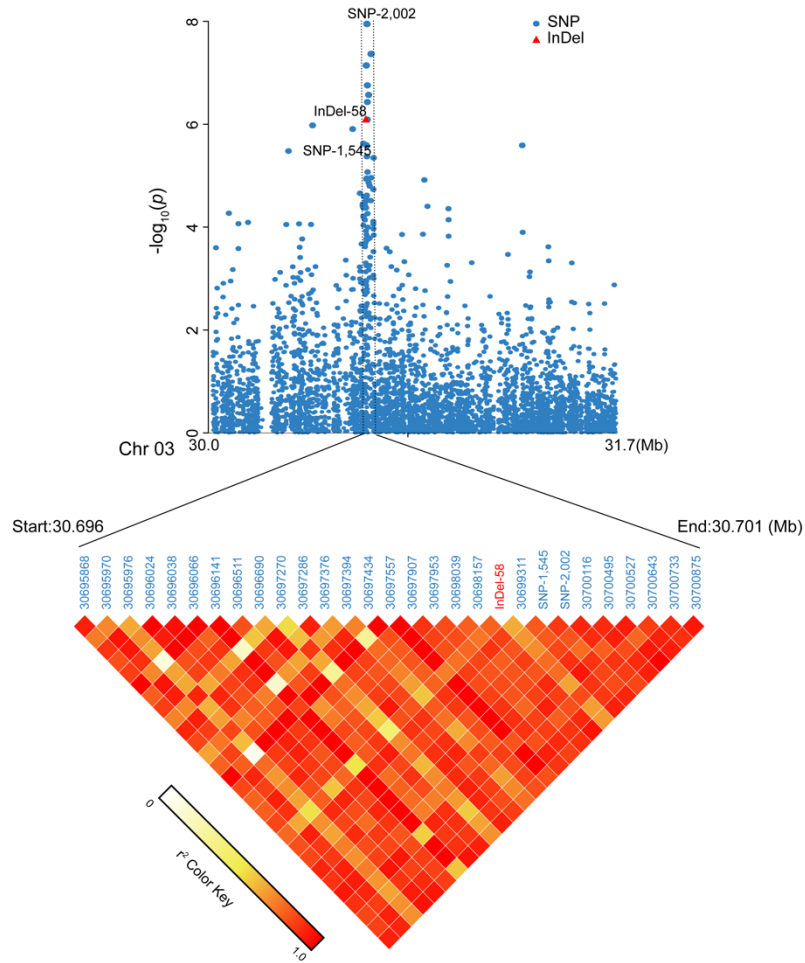
Supplementary Figure S9. MdNAC18.1 directly binds to the promoters of *MdPG1* and *MdACS1* in apples. **A)** Diagrams of the promoters of *MdPG1*. Fragment P1 and P2 contain one TTACGT motif, respectively. **B)** Diagrams of the promoters of *MdACS1*. Fragment P1 contains one TACGAG motif; fragment P2 contains one TACGTA motif. **C-D)** Binding of MdNAC18.1 to the *MdPG1* and *MdACS1* promoter as determined by ChIP-qPCR assay. Positions of fragments used for ChIP-qPCR are listed in **A-B**). EF1, and No-antibody serve as negative controls. Error bars indicate SD ($n = 3$). Student's two-tailed t test is used for statistical analyses, and statistically significant differences are indicated by *** $P < 0.001$. **E-F)** Binding of MdNAC18.1 to the promoter of *MdPG1* and *MdACS1* by electrophoretic mobility shift assay analysis. Mut represents the mutant probe. Binding sites in mutant probe were replaced with AAAAAA.



Supplementary Figure S10. Kompetitive Allele Specific PCR (KASP) analysis in the promoters of *MdNAC18.1* in 204 *Malus* accessions. The scatter plot with the axes X and Y represents the allelic discrimination of SNP-1,545 **A**) or SNP-2,002 **B**) genotypes. X- and Y-axes indicate the FAM and HEX fluorescence signals, respectively. The pink, blue, and orange dots represent the homozygous genotype of early-ripening, heterozygous genotype, and homozygous genotype of late-ripening from the mapping population of the 204 *Malus* cultivars, respectively.



Supplementary Figure S11. Detection of the InDel-58 in the *MdNAC18.1* promoters in 204 *Malus* accessions. M: DNA marker 500 (Takara, Japan).

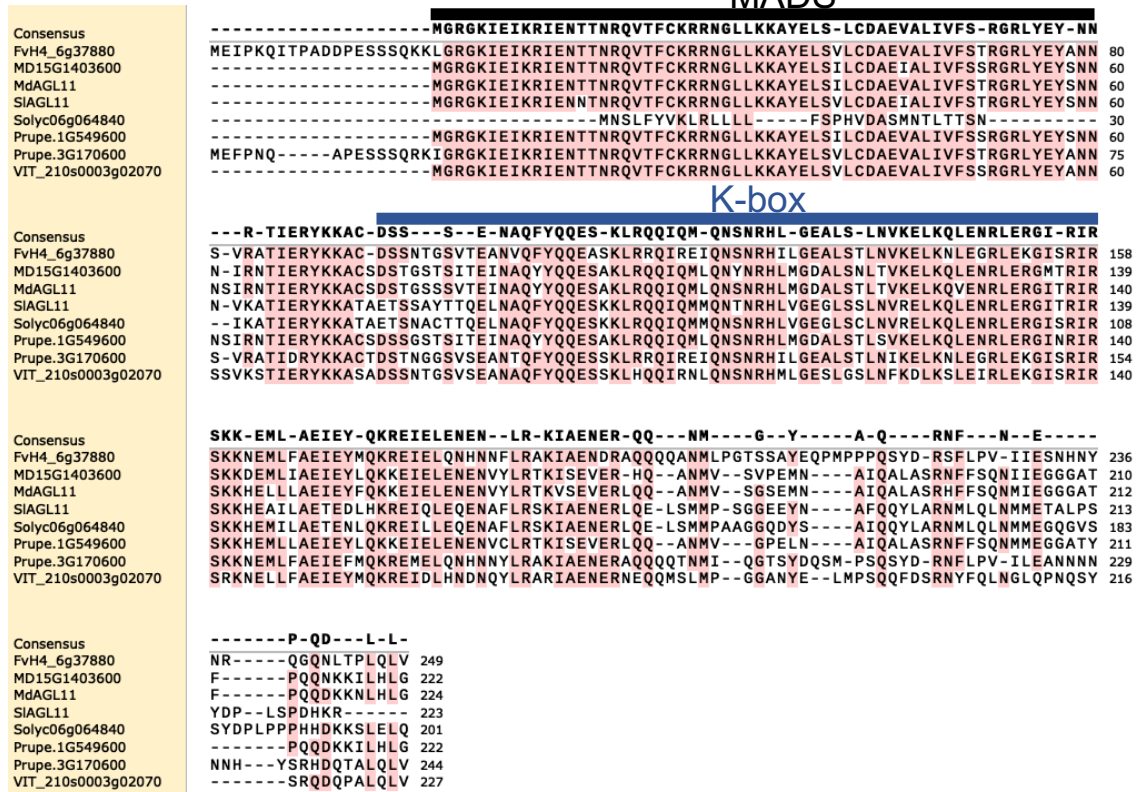


Supplementary Figure S12. Association analysis of the genetic variation with apple fruit ripening. Local Manhattan plot of fruit ripening using GWAS with SNPs in chromosome 03 (30.0 ~ 31.7 Mb) (top) and Linkage disequilibrium (LD) plot of the region surrounding the MdNAC18.1 (bottom). Dots denote SNPs, and triangles represent InDels. The p -value is shown on a $-\log_{10}$ scale. A 58-bp InDel variant (InDel-58) is highlighted in red. InDel-58, SNP-2,002, and SNP-1,545 within dotted lines are connected to the pairwise LD diagram. Chr 03, chromosome 03.

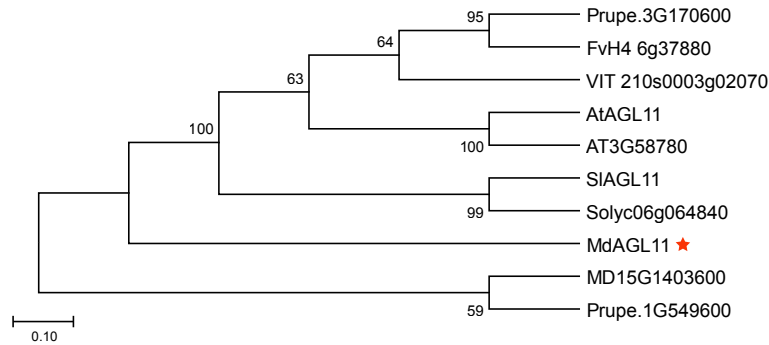


Supplementary Figure S13. Adjacent sequence analysis of InDel-58. Blue lines indicate CArG motif, pink lines indicate CAAT Box, and InDel-58 is marked in red or a dotted line.

A



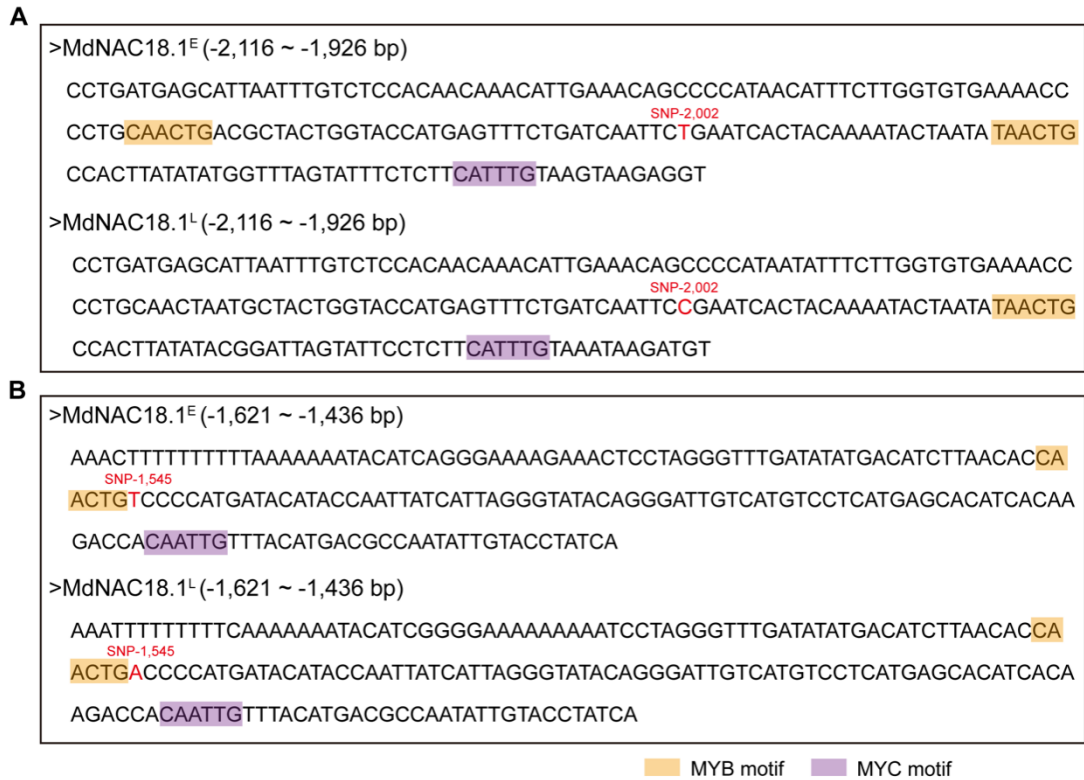
B



Supplementary Figure S14. Sequence alignment and phylogenetic analysis of MdAGL11. **A)** Sequence alignment of AGL proteins. Conserved domains are indicated by black and blue square. The yellow box indicates the name of the protein sequence. Pink highlighting indicates the same amino acids in the protein sequence alignment. **B)** Phylogenetic analysis of MdAGL11 proteins and their homologs. Evolutionary distances were estimated with a neighbor-joining algorithm. The numbers above the branches indicate the percentage from 1,000 bootstrap. The scale bar indicates the average number of amino acid substitutions per site. Accession numbers are as follows: *Solanum lycopersicum* (SIAGL11, Soly06g064840; Soly06g064840), *Prunus persica* (Prupe.3G170600; Prupe.1G549600), *Fragaria ananassa* (FvH4_6g37880), *Malus domestica* (MdAGL11, MD08G1216500; MD15G1403600), *Vitis vinifera* (VIT_210s0003g02070) and *Arabidopsis thaliana* (AtAGL11, AT4G609960; AT3G58780).

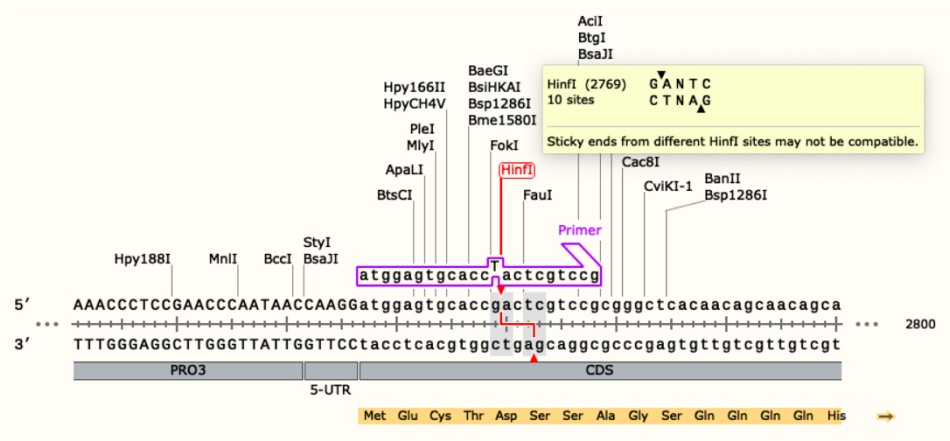


Supplementary Figure S15. Phylogenetic tree of *MdNAC18.1* branch. Genes marked in red labels and stars indicate functional NAC genes (regulate fruit ripening).

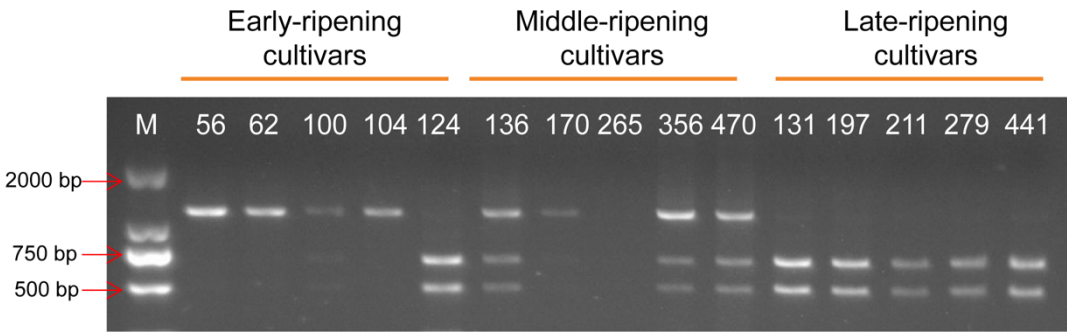


Supplementary Figure S16. Sequence analysis of MdNAC18.1^E and MdNAC18.1^L. **A)** Sequence analysis surrounding the SNP-2,002. **B)** Sequence analysis surrounding the SNP-1,545. The yellow boxes represent the MYB motif, the purple boxes represent the MYC motif.

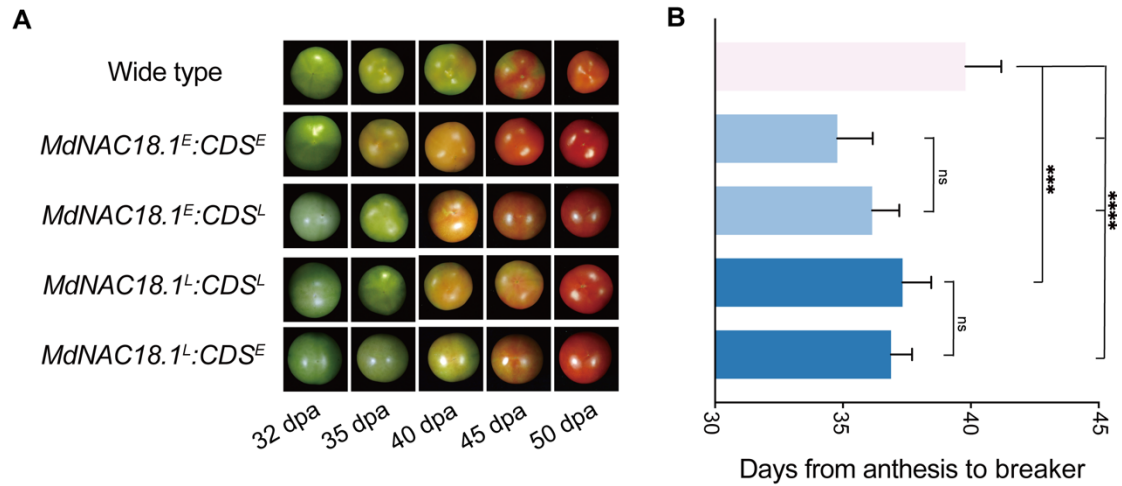
A



B



Supplementary Figure S17. D5Y SNP is associated with apple fruit ripening in 204 *Malus* accessions. **A)** Schematic diagram of CAPS site structure. **B)** Distribution of InDel-58 among *Malus* accessions detected by single enzyme digestion. M: DNA marker 2000 (Takara, Japan).



Supplementary Figure S18. Tomatoes with various transgenic initiated different fruits maturity. **A)** Representative fruits of wild type and various transgenic tomatoes at different stages. **B)** Days to color changes in different transgenic tomatoes and wild type. Error bars indicate SD ($n = 6$). Asterisks indicate the significant differences (One-way ANOVA with Dunnett's multiple comparisons test; *** $P < 0.001$, **** $P < 0.0001$, ns, not significant). dpa, days post-anthesis.